

From the Craig–Sulem Recurrence to a Learned Fourier-Filter Architecture

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Abstract

The Craig–Sulem expansion builds the water-wave Dirichlet–Neumann operator from Fourier filters and multiplication by surface-dependent fields. An exact polarization identity rewrites a symmetric pair of two-filter terms as a difference of branches that each use the same filter twice. This gives a mathematical motivation for our learned correction, while preserving an important distinction: higher-order terms also contain nested, nonlocal operations, so the finite branch sum is an approximation, not an exact reformulation of the recurrence.

1 The general recurrence

We use nondimensional spatial variables; network input/output normalization is a separate operation. Let x be periodic with period L , let $\eta(x)$ denote surface elevation, and let $h > 0$ denote the undisturbed depth. The potential ϕ satisfies Laplace’s equation between the impermeable bottom $z = -h$ and the surface $z = \eta(x)$, with surface value $\xi(x)$. We use the water-wave convention

$$G(\eta; h)\xi = (\phi_z - \eta_x \phi_x)|_{z=\eta(x)}. \quad (1)$$

This is the *unnormalized* normal derivative; with this convention, G is self-adjoint in $L^2(dx)$.

For wavenumbers $k = 2\pi\ell/L$, $\ell \in \mathbb{Z}$, define the known Fourier filters F_j by

$$F_j f = \mathcal{F}^{-1}[s_j(k, h) \mathcal{F} f], \quad s_j(k, h) = \begin{cases} |k|^j, & j \text{ even}, \\ |k|^j \tanh(h|k|), & j \text{ odd}. \end{cases} \quad (2)$$

Here $s_0 = 1$, including at $k = 0$. In particular, $F_0 = I$, $F_1 = G_0$, and $F_2 = -\partial_x^2$.

Write $G(\eta; h) = \sum_{n \geq 0} G_n(\eta; h)$, where G_n is homogeneous of degree n in η , and set $g_n = G_n(\eta; h)\xi$. A compact adjoint form of the Craig–Sulem recurrence is

$$g_0 = F_1 \xi, \quad (3)$$

$$g_n = -F_{n-1} \left[\partial_x \left(\frac{\eta^n}{n!} \partial_x \xi \right) \right] - \sum_{j=1}^n F_j \left[\frac{\eta^j}{j!} g_{n-j} \right], \quad n \geq 1. \quad (4)$$

All products inside brackets are pointwise products in physical space. The finite-depth recurrence is given in parity-split form by Nicholls [1]; Appendix A derives the ordering used here. We use it as a local shape expansion about $\eta = 0$, not as a claim of convergence for arbitrary steep surfaces.

The structural message is simple: *Fourier filters handle wavenumber and depth; multiplication by powers of η introduces the surface geometry*. Substituting lower-order terms repeatedly produces sums of alternating chains of these operations.

2 What the first corrections reveal

The first two corrections are [2]

$$G_1(\eta; h)\xi = -G_0[\eta G_0\xi] - \partial_x[\eta \partial_x \xi], \quad (5)$$

$$G_2(\eta; h)\xi = G_0[\eta G_0[\eta G_0\xi]] + \frac{1}{2}G_0[\eta^2 \partial_x^2 \xi] + \frac{1}{2}\partial_x^2[\eta^2 G_0\xi]. \quad (6)$$

Depth dependence in G_0 is suppressed to keep the expressions readable.

The last two terms in (6) use the same physical-space weight η^2 , with the inner and outer filters exchanged. Together they form a self-adjoint pair. The first term is different: it contains three filters and two surface multiplications. It will identify the approximation made by the architecture.

3 Why use the same filter twice?

For a Fourier coefficient vector m , define its action explicitly:

$$\text{filter}(m, f) = \text{IFFT}[m \text{ FFT}(f)]. \quad (7)$$

Thus m is a vector of filter coefficients, not an operator or a function of the input field f . For compact algebra, write $M_m f = \text{filter}(m, f)$. Let $A_a f = a(x)f(x)$ denote multiplication by a real spatial field a .

For two filter vectors p and q , the identity

$$\boxed{\frac{1}{2}(M_p A_a M_q + M_q A_a M_p) = \frac{1}{4}M_{p+q} A_a M_{p+q} - \frac{1}{4}M_{p-q} A_a M_{p-q}} \quad (8)$$

is exact. Expanding the right-hand side cancels the $M_p A_a M_p$ and $M_q A_a M_q$ terms and leaves the two cross terms. No multiplication operator has been commuted past a Fourier filter.

Consequently, a symmetric pair with different inner and outer filters is exactly a sum of two *tied-filter* branches: each branch uses one filter on both sides of a spatial multiplication. Their spatial weights are $a/4$ and $-a/4$. For real, even Fourier symbols and real a , each branch is self-adjoint and maps real fields to real fields.

For example, the last two terms of (6) have

$$p(k) = |k| \tanh(h|k|), \quad q(k) = -k^2, \quad a(x) = \eta(x)^2. \quad (9)$$

Equation (8) therefore applies directly to this pair. The identity concerns the filter operations themselves; representing the required coefficient functions with finite neural networks remains a separate approximation.

4 The resulting architectural motivation

The Craig–Sulem recurrence combines Fourier multipliers with multiplication by powers of the surface elevation. We retain these two ingredients in our branches: we learn the Fourier multipliers and learn the spatial multiplication coefficients as functions of explicit powers of η , augmented by derivatives and transforms of the surface.

Before network input/output normalization, the model has the form

$$G_{\text{learned}}(\eta; h)\xi = G_0(h)\xi + G_1(\eta; h)\xi + \frac{1}{\sqrt{B}} \sum_{b=1}^B \text{filter}(m_b, a_b[\eta] \text{ filter}(m_b, \xi)). \quad (10)$$

Here B is the number of branches, $m_b(k, h)$ is a learned real, even Fourier symbol, and $a_b[\eta](x)$ is a learned real spatial field computed from the surface features specified below. The brackets emphasize that a_b depends on derivatives and transforms of η as well as its value at x . Neither m_b nor a_b depends on ξ .

Explicit powers in the surface network. Let $\tilde{\eta} = \eta/s_\eta$, where $s_\eta > 0$ is the fixed training-set normalization scale. The implementation constructs $\tilde{\eta}^p$ for $p = 1, \dots, P$, with P set by `cs_n_polys`. The current C27 configuration uses $P = 3$ and enables all four derivative/transform channels, giving the seven-channel input

$$E_0[\eta](x) = [\tilde{\eta}, \tilde{\eta}^2, \tilde{\eta}^3, \partial_x \tilde{\eta}, \partial_x^2 \tilde{\eta}, |D|^{1/2} \tilde{\eta}, \mathcal{H} \tilde{\eta}]^\top(x). \quad (11)$$

Here $|D|^{1/2}$ and the Hilbert transform $\mathcal{H}$ have Fourier symbols $|k|^{1/2}$ and $-i \operatorname{sgn}(k)$, respectively. Two shared, bias-free pointwise dense layers with GELU activations map these inputs to a latent vector $z_\theta[\eta](x)$, with 320 channels in C27. A bias-free linear head for each branch produces

$$a_b[\eta](x) = w_b^\top [z_\theta[\eta](x) \odot \tanh(z_\theta[\eta](x))], \quad (12)$$

where $\odot$ denotes componentwise multiplication. Thus the spatial coefficients are explicitly learned functions of the first three surface powers and the auxiliary channels. The polynomial inputs supply coordinates motivated by the recurrence; nonlinear mixing allows dependence beyond cubic order. The separate bias-free construction and nonlinear lift enforce the small-amplitude order described next.

For fixed η and h , each branch is linear in ξ and self-adjoint:

$$(M_{m_b} A_{a_b} M_{m_b})^* = M_{m_b} A_{a_b} M_{m_b}. \quad (13)$$

The current surface network also ensures $a_b[\varepsilon\eta] = O(\varepsilon^2)$ as $\varepsilon \rightarrow 0$: with the normalization scale held fixed, $z_\theta[\varepsilon\eta] = O(\varepsilon)$, and the lift in (12) satisfies $z \tanh z = O(z^2)$. The learned correction therefore starts at second order, leaving the analytic zeroth- and first-order terms unchanged.

Signed weights are intentional. For real ξ ,

$$\langle \xi, M_m A_a M_m \xi \rangle = \int_0^L a(x) |M_m \xi(x)|^2 dx. \quad (14)$$

Nonnegative a would make a branch positive semidefinite, but would exclude the negative branch used in (8). The full physical DNO is positive semidefinite; its individual Taylor corrections need not be. Self-adjointness alone does not enforce positivity of the learned total.

The approximation is the shallow branch sum. The nested term $G_0[\eta G_0[\eta G_0\xi]]$ in (6) does not generally reduce to $M_m[a M_m\xi]$: the middle operation $f \mapsto \eta G_0[\eta f]$ is nonlocal, not multiplication by one fixed field. Our finite sum approximates such effects instead of explicitly executing the full recurrence. Neither the recurrence nor the polarization identity establishes a sufficient branch count, universal expressivity, constant-nullspace preservation, or rollout stability.

Implementation scope. The expression above describes the analytic structure. In `models/dno-net/dno_net_v2.py`, normalization is handled separately, depth is replaced by $\min(h, 5)$, and the baseline products use the model’s collocation grid. Thus “analytic baseline” means the prescribed $G_0 + G_1$ formulas, not an exact continuum evaluation for every depth and discretization.

Suggested manuscript wording. We retain the analytic $G_0 + G_1$ baseline and approximate the higher-order correction using signed, self-adjoint branches composed of Fourier filters and surface-dependent multiplication. The Craig–Sulem recurrence motivates both ingredients: we learn the Fourier multipliers and learn the spatial multiplication coefficients from the normalized surface powers $\tilde{\eta}$, $\tilde{\eta}^2$, and $\tilde{\eta}^3$, together with derivative and transform channels. An exact polarization identity motivates tying the inner and outer filters, while a separate bias-free network and quadratic lift ensure that the learned correction begins at second order. The finite branch sum is a structured approximation to the remaining, generally nested terms.

A Derivation of the compact recurrence

For the formal expansion, parameterize a harmonic field by its flat-level trace f . Its Fourier coefficients satisfy

$$\widehat{\phi}(k, z) = \frac{\cosh((z + h)|k|)}{\cosh(h|k|)} \widehat{f}(k). \quad (15)$$

At $z = 0$, the j th vertical derivative is $F_j f$, which explains the even/odd symbols in (2).

Let E_j mean multiplication by $\eta^j/j!$, with $E_0 = I$. Taylor expansion of the surface trace gives

$$\xi = \sum_{j \geq 0} E_j F_j f. \quad (16)$$

The order- n contribution to its unnormalized normal derivative, for $n \geq 1$, is

$$E_n F_{n+1} f - \eta_x E_{n-1} \partial_x F_{n-1} f = -\partial_x E_n \partial_x F_{n-1} f. \quad (17)$$

This uses $F_{n+1} = -\partial_x^2 F_{n-1}$ and $\partial_x(\eta^n/n!) = \eta_x \eta^{n-1}/(n-1)!$.

Apply $G(\eta; h) = \sum_{n \geq 0} G_n$ to the expanded trace and match powers of η . Since the flat-level trace f is arbitrary, the result is the operator recurrence

$$G_n = -\partial_x E_n \partial_x F_{n-1} - \sum_{j=1}^n G_{n-j} E_j F_j, \quad G_0 = F_1. \quad (18)$$

Every Taylor coefficient G_n is self-adjoint, as follows by differentiating the self-adjoint family $G(\varepsilon\eta; h)$ at $\varepsilon = 0$. Taking adjoints, using $E_j^* = E_j$, $F_j^* = F_j$, and $\partial_x^* = -\partial_x$, gives

$$G_n = -F_{n-1} \partial_x E_n \partial_x - \sum_{j=1}^n F_j E_j G_{n-j}. \quad (19)$$

Applying this identity to ξ yields (4). It is the ordering implemented, with even and odd orders separated, in `solver/solvers/dno_series_jax.py`.

References

- [1] David P. Nicholls. Traveling Water Waves: Spectral Continuation Methods with Parallel Implementation. *Journal of Computational Physics*, 143:224–240, 1998. See Section 2.2, especially equations (2.7)–(2.11).
- [2] Jon Wilkening and Vishal Vasan. Comparison of five methods of computing the Dirichlet–Neumann operator for the water wave problem. *Contemporary Mathematics*, 635:175–210, 2015. Preprint: arXiv:1406.5226, 2014. See Section 2.1.